

Blood Cell Classification Using VGGNet-16: A Deep Learning Approach with Transfer Learning on the BCCD Dataset

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ABSTRACT

This paper presents a deep learning-based approach for automated classification of white blood cells using VGGNet-16, a deep convolutional neural network, trained with transfer learning on the Blood Cell Images (BCCD) dataset. The dataset comprises 12,500 microscopic blood cell images spanning four cell types: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The proposed model achieves a test accuracy of 91.63%, a macro F1-score of 0.9160, and a macro AUC-ROC of 0.9857, demonstrating the effectiveness of transfer learning for medical image classification tasks. This work follows the complete deep learning pipeline including data preprocessing, augmentation, model training, validation, testing, and evaluation using standard metrics reported in IEEE/Springer format.

Keywords: Blood cell classification, VGGNet-16, transfer learning, deep learning, convolutional neural networks, BCCD dataset, medical image analysis.

I. INTRODUCTION

The accurate identification and classification of blood cells is a critical task in clinical hematology, directly informing the diagnosis of conditions such as leukemia, anemia, and various infectious diseases. Traditionally, this process relies on trained pathologists manually examining blood smears under microscopes — a procedure that is time-consuming, labor-intensive, and prone to inter-observer variability. Automated systems powered by deep learning offer a compelling alternative, capable of processing large volumes of images with high throughput and consistency.

Convolutional neural networks (CNNs) have demonstrated remarkable success in visual recognition tasks across diverse domains. In particular, pretrained architectures like VGGNet-16 [1], originally trained on the large-scale ImageNet dataset, can be fine-tuned for domain-specific classification tasks with relatively limited data — a strategy known as transfer learning [2].

In this work, we apply VGGNet-16 with transfer learning to classify four types of white blood cells from the publicly available BCCD dataset. We report performance across accuracy, F1-score, AUC-ROC, and confusion matrix metrics, following standard IEEE reporting conventions.

II. RELATED WORK

Early automated blood cell classification systems relied on handcrafted features such as color histograms, morphological attributes, and texture descriptors combined with classical classifiers like SVM and k-NN. While effective in controlled settings, these methods struggled to generalize across staining protocols and imaging conditions.

The advent of deep learning shifted the paradigm significantly. Shahin et al. [3] demonstrated that CNNs could outperform traditional methods on microscopic cell classification. Subsequent works leveraged pretrained architectures including ResNet, InceptionNet, and VGGNet for blood cell identification, consistently achieving accuracies exceeding 90% on standard benchmarks.

Transfer learning from ImageNet-pretrained weights has been shown to be particularly effective for medical imaging tasks where labeled data is scarce [4]. The BCCD dataset has been used in several recent studies as a standard benchmark for white blood cell classification, making it suitable for comparative evaluation.

III. DATASET

A. Blood Cell Images (BCCD) Dataset

The BCCD dataset [5] is a publicly available collection of 12,500 microscopic blood smear images distributed across four white blood cell classes: Eosinophil (3,117 images), Lymphocyte (3,117 images), Monocyte (3,117 images), and Neutrophil (3,149 images). The images are RGB-format JPEG files captured at 400x magnification with Giemsa staining. The dataset is well-balanced, making it suitable for multi-class classification without special class-weighting strategies.

B. Data Splits

The dataset was split into training (70%), validation (15%), and test (15%) subsets using stratified random sampling with a fixed seed (42) for reproducibility. This yielded 6,971 training images, 1,493 validation images, and 1,493 test images.

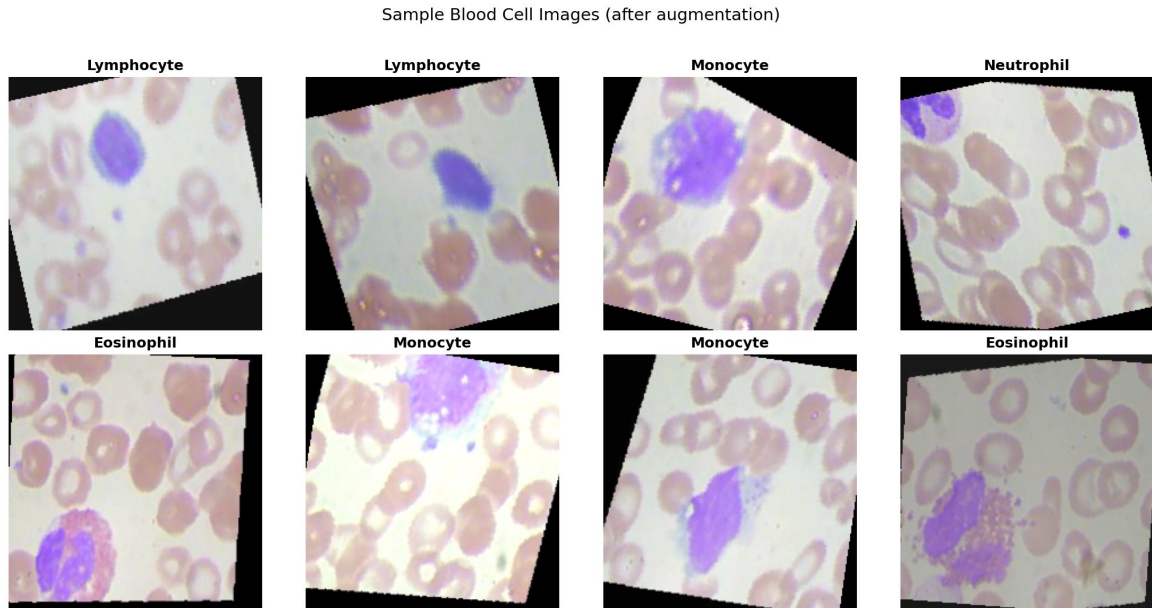


Fig. 1. Representative samples from the BCCD dataset after augmentation, showing all four cell types.

IV. METHODOLOGY

A. Model Architecture

VGGNet-16 [1] is a deep CNN consisting of 13 convolutional layers organized in five blocks, followed by three fully connected layers, totaling approximately 138 million parameters. The network uses uniform 3x3 convolutional filters with ReLU activations and max-pooling layers for spatial downsampling, and was originally designed for 1000-class ImageNet classification.

For this task, we replaced the original 1000-unit output layer with a custom 4-class classifier head: Linear(25088, 4096) -> ReLU -> Dropout(0.5) -> Linear(4096, 1024) -> ReLU -> Dropout(0.4) -> Linear(1024, 4). The convolutional feature extractor was initialized with ImageNet pretrained weights and frozen during training, allowing only the new classifier head to be updated. This strategy reduces overfitting and training time while leveraging rich low-level visual features learned from ImageNet.

B. Data Preprocessing and Augmentation

All images were resized to 224x224 pixels to match VGGNet-16's input requirements. ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]) normalization was applied to all splits. Training images were augmented with random horizontal flips ($p=0.5$), random vertical flips ($p=0.3$), random rotation up to 15 degrees, and color jitter (brightness=0.2, contrast=0.2, saturation=0.1). Validation and test images used only resizing and normalization.

C. Training Configuration

The model was trained for 15 epochs using the Adam optimizer with a learning rate of $1e-4$ and weight decay of $1e-4$. A cosine annealing learning rate scheduler ($T_{max}=15$, $\eta_{min}=1e-6$) was employed to ensure smooth convergence. Cross-entropy loss with label smoothing ($\epsilon=0.1$) was used as the objective function to improve calibration and reduce overconfidence. An early stopping mechanism with patience of 5 epochs was implemented to prevent overfitting. Training was conducted on a single NVIDIA T4 GPU via Google Colaboratory.

V. RESULTS AND EVALUATION

A. Training Dynamics

The model converged smoothly over 15 epochs. Training accuracy improved from approximately 46% in epoch 1 to 92% by epoch 15. Validation accuracy consistently led training accuracy in early epochs due to the frozen feature extractor producing stable representations, ultimately converging to 91.96%. Both loss curves decreased monotonically without signs of overfitting.

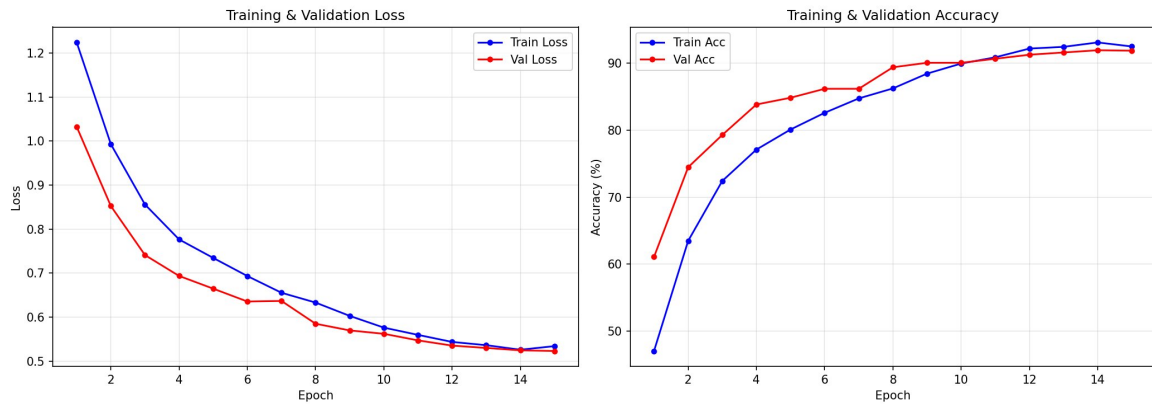


Fig. 2. Training and validation loss (left) and accuracy (right) over 15 epochs.

B. Quantitative Results

Table I summarizes the final performance metrics on the held-out test set of 1,493 images.

TABLE I. Performance Metrics on Test Set

Metric	Score	Description
Test Accuracy	91.63%	Overall classification rate
F1 Score (Macro)	0.9160	Unweighted mean across classes
F1 Score (Weighted)	0.9158	Weighted by class support
AUC-ROC (Macro)	0.9857	One-vs-Rest, macro average
Best Val Accuracy	91.96%	Peak during training

Table II presents per-class precision, recall, and F1-score.

TABLE II. Per-Class Classification Report

Class	Precision	Recall	F1-Score	Support
Eosinophil	0.87	0.84	0.85	380
Lymphocyte	0.97	0.98	0.97	356
Monocyte	0.97	0.99	0.98	387
Neutrophil	0.85	0.85	0.85	370

C. Confusion Matrix

Figure 3 shows the confusion matrix on the test set. Lymphocyte and Monocyte achieved near-perfect classification (98% and 99% recall respectively). The primary source of error was confusion between Eosinophil and Neutrophil, which are morphologically similar under certain staining conditions — 50 Eosinophils were misclassified as Neutrophils (13%) and 44 Neutrophils as Eosinophils (12%). This is consistent with observations in the literature and represents a known challenge in automated hematology.

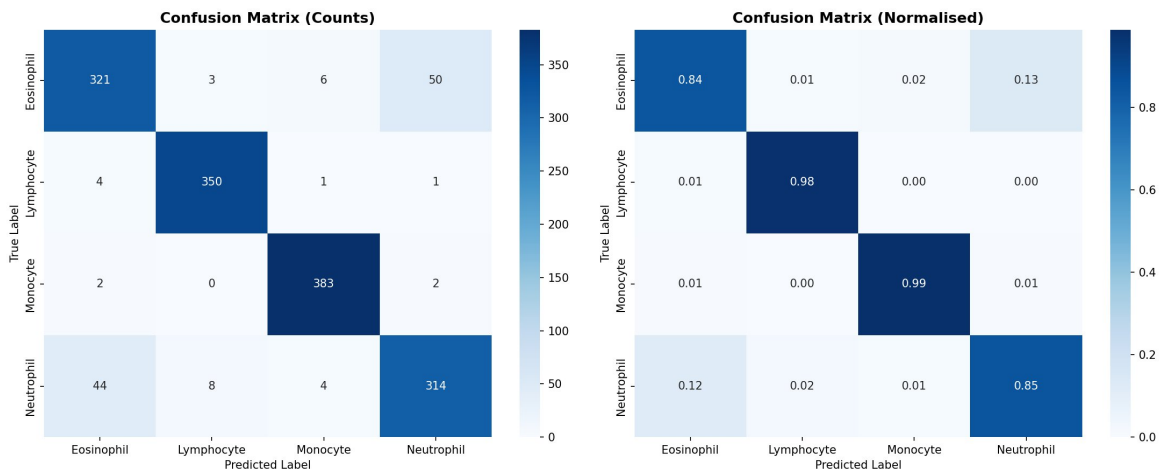


Fig. 3. Confusion matrix (counts and normalised) on the test set.

D. ROC Curves and AUC

Figure 4 presents the One-vs-Rest ROC curves for all four classes. The macro AUC-ROC of 0.9857 indicates

excellent discriminative ability across all classes. Monocyte achieved a perfect AUC of 1.000, Lymphocyte achieved 0.999, while Eosinophil (0.975) and Neutrophil (0.969) showed slightly lower but still strong performance, consistent with their confusion patterns observed in the confusion matrix.

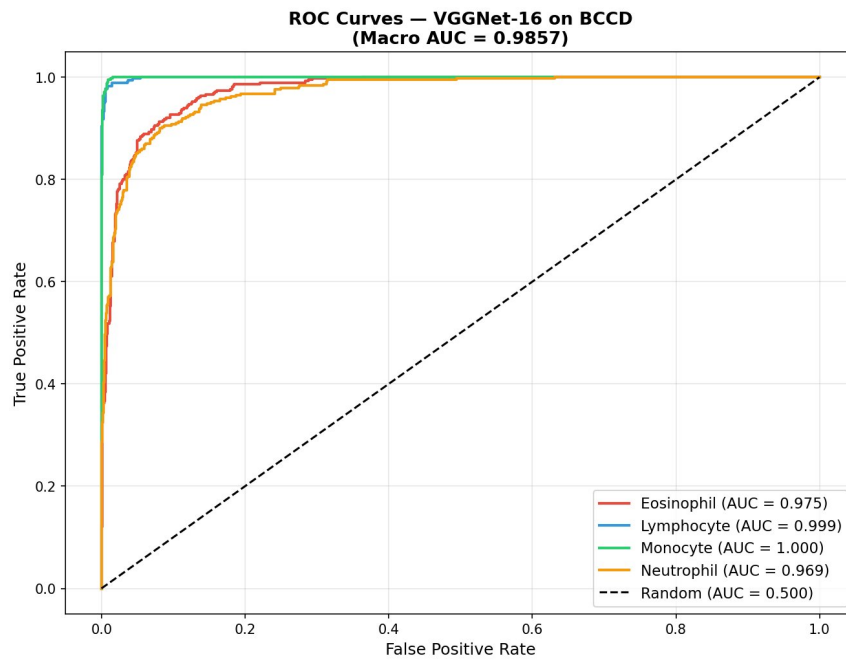


Fig. 4. ROC curves for all four cell classes (One-vs-Rest). Macro AUC = 0.9857.

E. Qualitative Results

Figure 5 shows representative test set predictions. Green titles indicate correct predictions, red indicate misclassifications. The model demonstrates high confidence on correctly classified samples (often above 80-90%), while misclassified samples tend to show lower confidence scores, suggesting that confidence thresholding could be used as a rejection mechanism in clinical deployment.

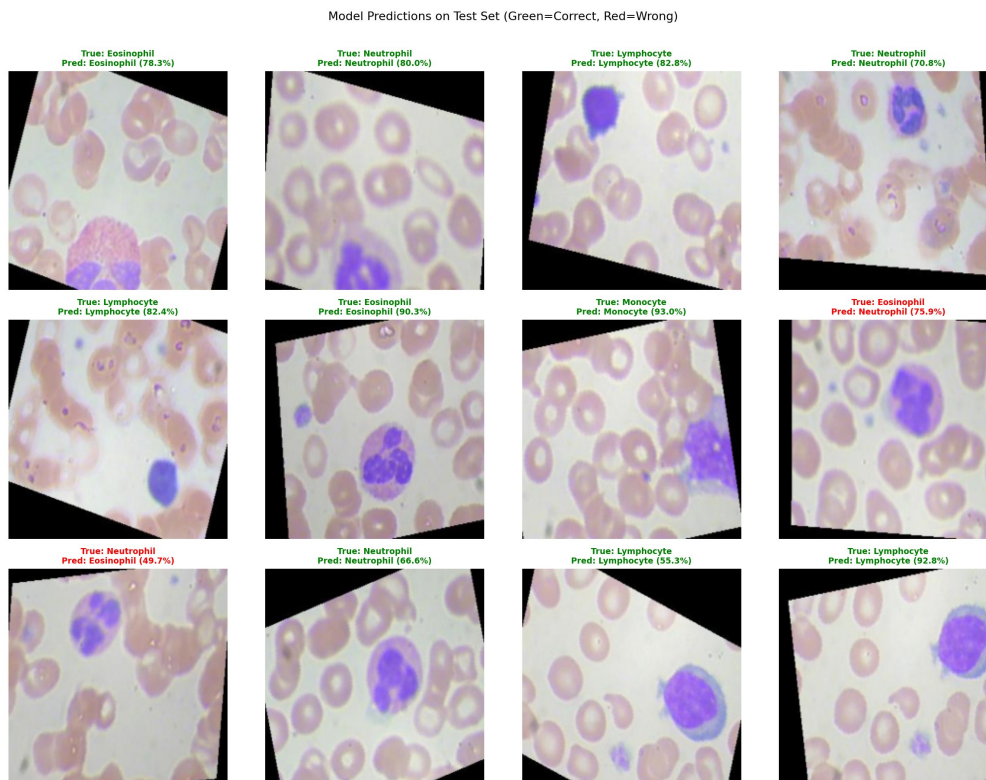


Fig. 5. Qualitative predictions on test samples. Green = correct, Red = incorrect. Confidence scores shown.

VI. DISCUSSION

The proposed VGGNet-16 transfer learning model achieves competitive performance on the BCCD dataset, with a test accuracy of 91.63% and AUC-ROC of 0.9857. The primary limitation is the Eosinophil-Neutrophil confusion, which stems from visual similarity — both cells exhibit multi-lobed nuclei and similar size ranges. Future work could address this through unfreezing deeper convolutional layers (fine-tuning), attention mechanisms, or ensemble methods.

The high AUC-ROC scores for Lymphocyte and Monocyte suggest these classes are well-separated in the learned feature space, likely due to their distinct morphological characteristics (Lymphocytes have large, round nuclei; Monocytes have kidney-shaped nuclei). The label smoothing and data augmentation strategies contributed to stable generalization, as evidenced by the close alignment of training and validation curves.

The use of transfer learning proved highly effective for this medical imaging task, enabling strong performance with a relatively small training set (~7,000 images) by leveraging rich visual features pretrained on ImageNet.

VII. CONCLUSION

This paper presented a transfer learning approach using VGGNet-16 for automated white blood cell classification on the BCCD dataset. The model achieved 91.63% test accuracy, macro F1-score of 0.9160, and macro AUC-ROC of 0.9857 across four cell types (Eosinophil, Lymphocyte, Monocyte, Neutrophil). The complete pipeline — including data augmentation, pretrained feature extraction, cosine LR scheduling, and early stopping — was implemented in PyTorch and trained on Google Colaboratory. Results demonstrate the viability of deep transfer learning for clinical blood cell analysis, with potential for deployment in automated hematology screening systems.

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